

SORTING

PROBLEM 2

QUESTION:

DS	Session	Sorting	Question Information	Level 1	Challenge 12
Question description Sajid is an third year student in a reputed institution. Although he scored well in many subjects, he did not an expert in computer programming languages. But Sajid's computer examination is scheduled for next week. As per the blueprint, many questions would come from the sorting topic. He collected previous year's questions. one of the repeated questions is to sort the given set of numbers using Selection Sort The first line of the input contains the number of elements N, the second line of the input contains the numbers A_i to be sorted. In the output print the the final sorted array in the given format. Can you help him ? Constraints $1 \leq N \leq 10^5$ $1 \leq A_i \leq 10^9$ Input: The first line of the input contains the number of elements the second line of the input contains the numbers to be sorted. Output: print the the final sorted array in the given format.					

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
5 15 17 11 25 1	6 65 45 27 21 25 11
EXPECTED OUTPUT	EXPECTED OUTPUT
1 11 15 17 25	11 21 25 27 45 65

Mandatory Test Cases

Test Case 1	Test Case 2	Test Case 3
KEYWORD	KEYWORD	KEYWORD
void selectionSort(int arr[],int n)	void swap(int *xp,int *yp)	void printArray(int arr[],int size)

Complexity Test Cases

Test Case 1	Test Case 2	Test Case 3
CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
2	275	47

SOURCE CODE:

```
#include <stdio.h>

void perform_selection_sort(int data[], int length) {
    for (int step = 0; step < length - 1; step++) {
        int best_pos = step;
        for (int probe = step + 1; probe < length; probe++) {
            if (data[probe] < data[best_pos]) best_pos = probe;
        }
        int temp = data[step];
        data[step] = data[best_pos];
        data[best_pos] = temp;
    }
}

int main() {
    int total;
    if (scanf("%d", &total) != 1) return 0;

    int numbers[total];
    for (int i = 0; i < total; i++) scanf("%d", &numbers[i]);

    perform_selection_sort(numbers, total);

    for (int i = 0; i < total; i++) {
        printf("%d%s", numbers[i], (i == total - 1) ? "" : " ");
    }
    printf("\n");
    return 0;
}
```

OUTPUT:

```
6
65 45 27 21 25 11
11 21 25 27 45 65
```

```
5
15 17 11 25 1
1 11 15 17 25
```